Topic Name:
Introduction to Deep Learning

Deep learning refers sometimes to training large neural networks.

Common Applications of Deep Learning

Input(x)	Output (y)	Application
Home features	Price	Real Estate
Ad, user info	Click on ad? (0/1)	Online Advertising
Image	Object (1,…,1000)	Photo tagging
Audio	Text transcript	Speech recognition
English	Chinese	Machine translation
Image, Radar info	Position of other cars	Autonomous driving

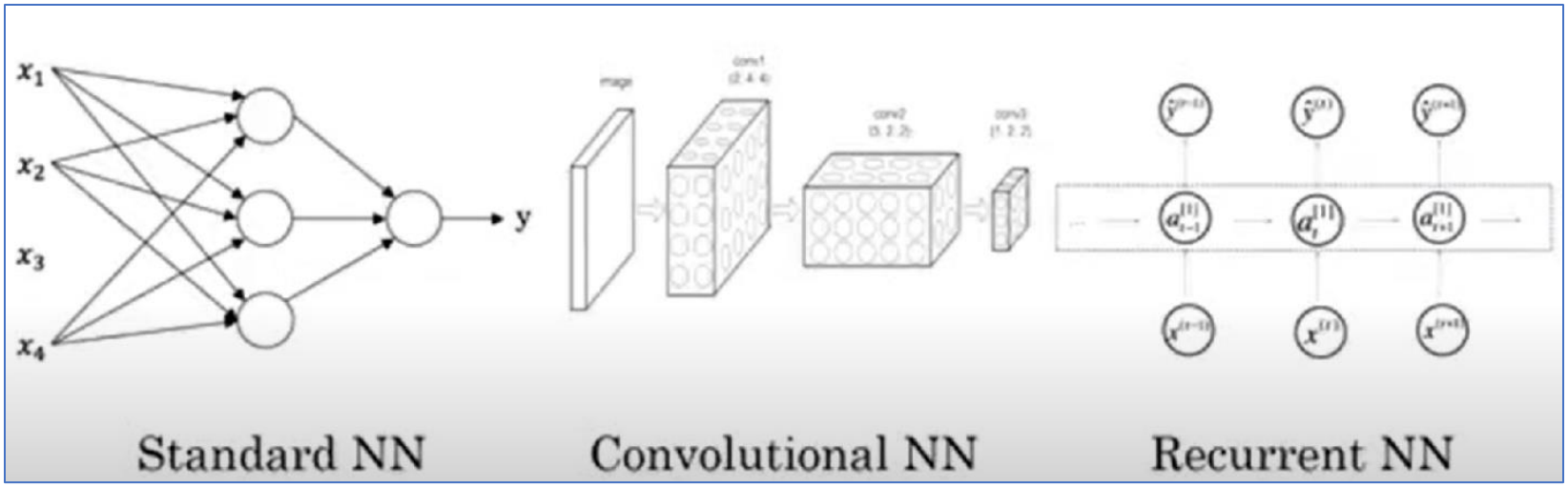
- There are slightly 20 different types of NN which are useful for different applications.

For example:

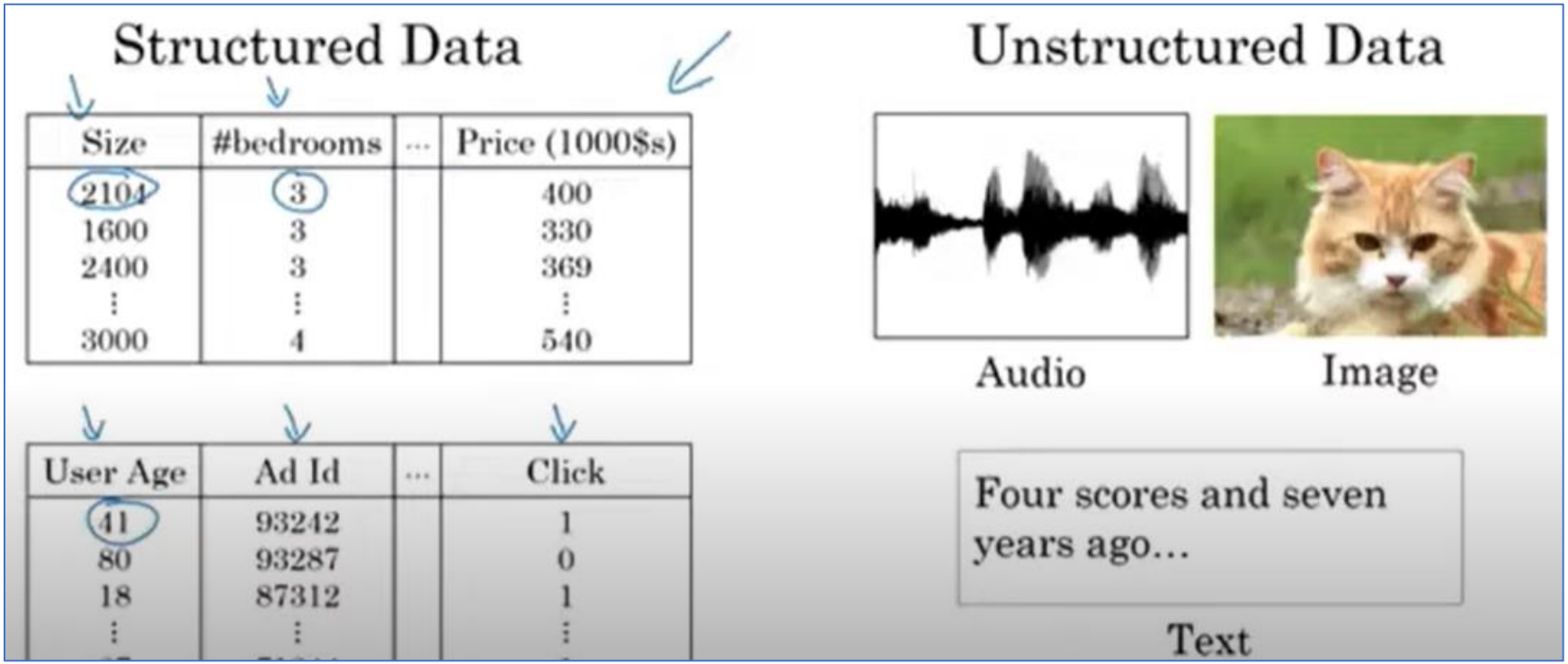
- Real estate and online advertising (standard NN)
- Images (CNN)
- Speech recognition and machine translation (RNN)
- Autonomous driving (complex hybrid NN)

CNN: Convolutional Neural Networks

RNN: Recurrent Neural Networks

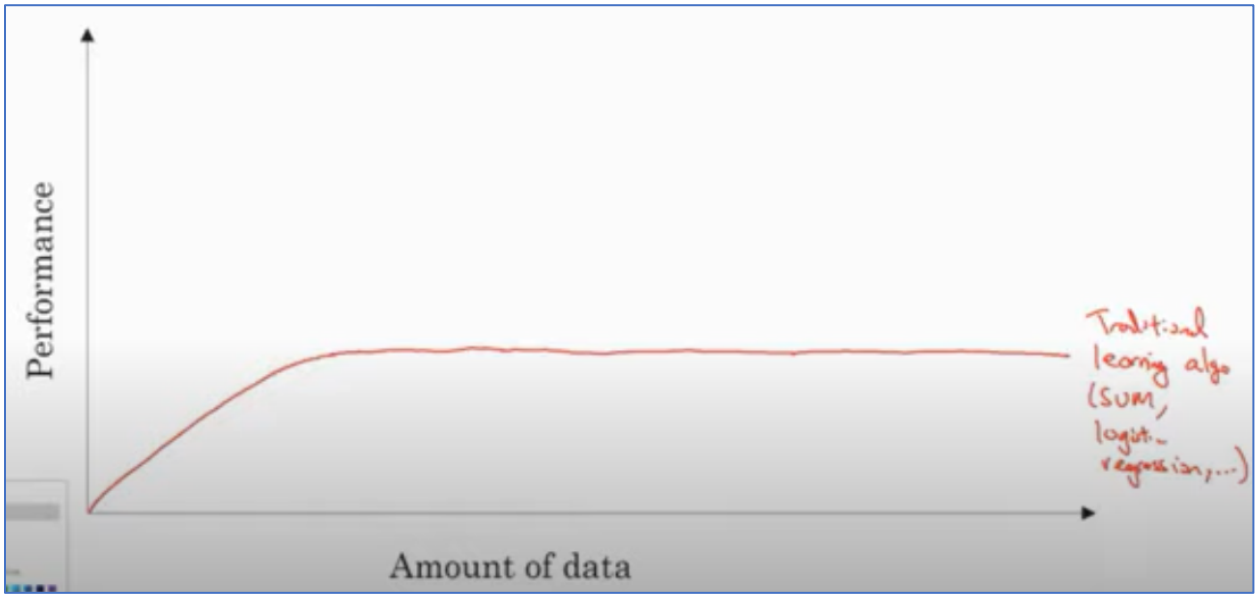

x_1
x_2
x_3
x_4
y
Standard NN
Convolutional NN
Recurrent NN

Structured Data vs. Unstructured Data

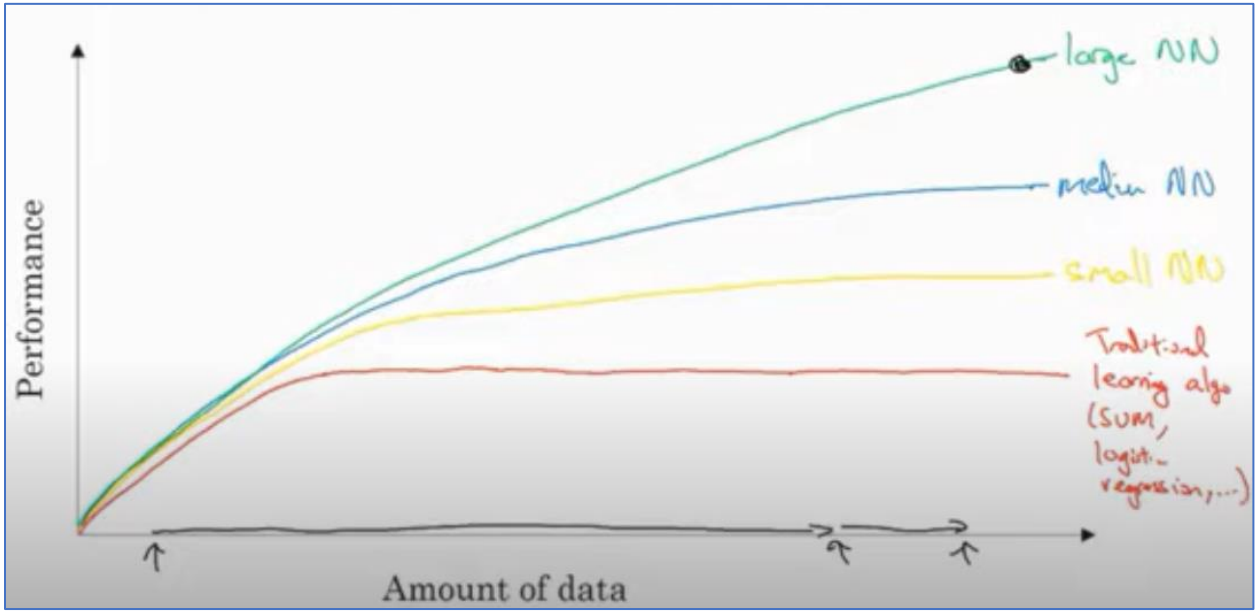


Historically it is much harder for computers to make sense of unstructured data compared to structured data.

The ideas that stand behind using deep learning these days



Traditional machine learning algorithms


Performance
large NN
medium NN
small NN
Traditional learning algo (SVM, logistic regression, ...)
Amount of data

Many reasons force us to go from a relatively small amount of data to having a large amount of data:

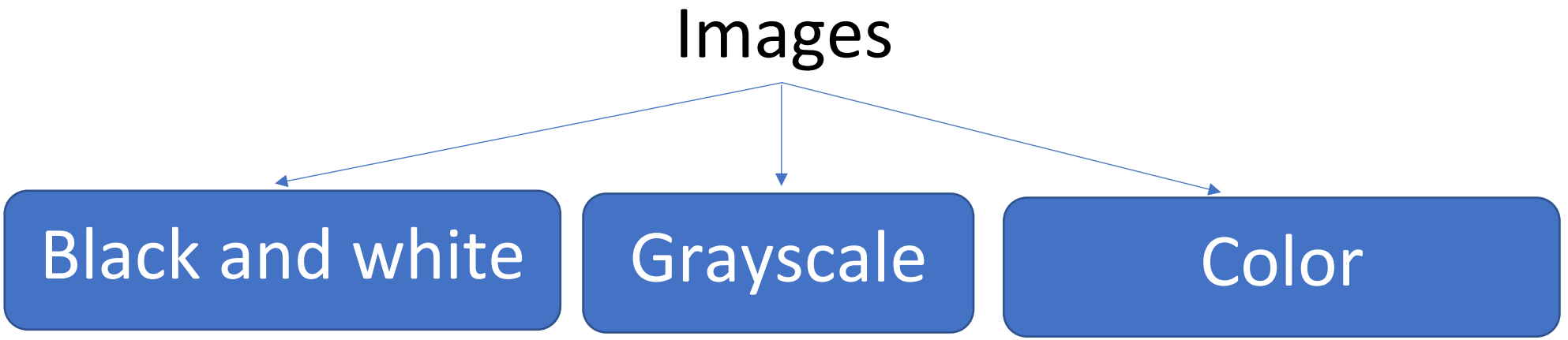
- Digitization of a society (we spend so much time on the computers, websites, mobile apps and activities on digital devices).
- The rise of inexpensive cameras built into our cell phones.
- All sorts of sensors in the Internet of Things.

Two observations if you want to hit this very high level of performance:

- you need to be able to train a big enough neural network in order to take advantage of the huge amount of data (high performance computing).
- you need a lot of data

so we often say that scale has been driving deep learning progress and by scale we mean both the size of the neural network and the amount of data.

Classification of Images using NN









Examples

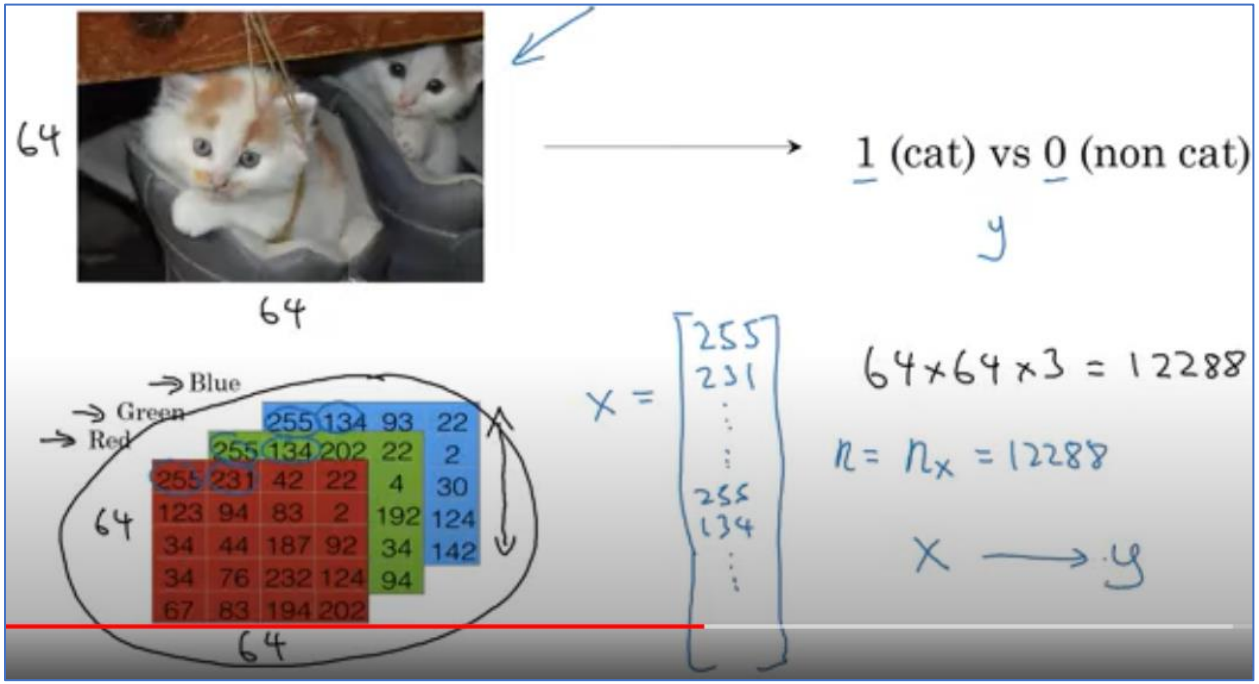
Black and white (Zero and one)

0	0	0	0	0	0	0
1	1	1	1	1	1	1
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0
0	0	0	1	0	0	0

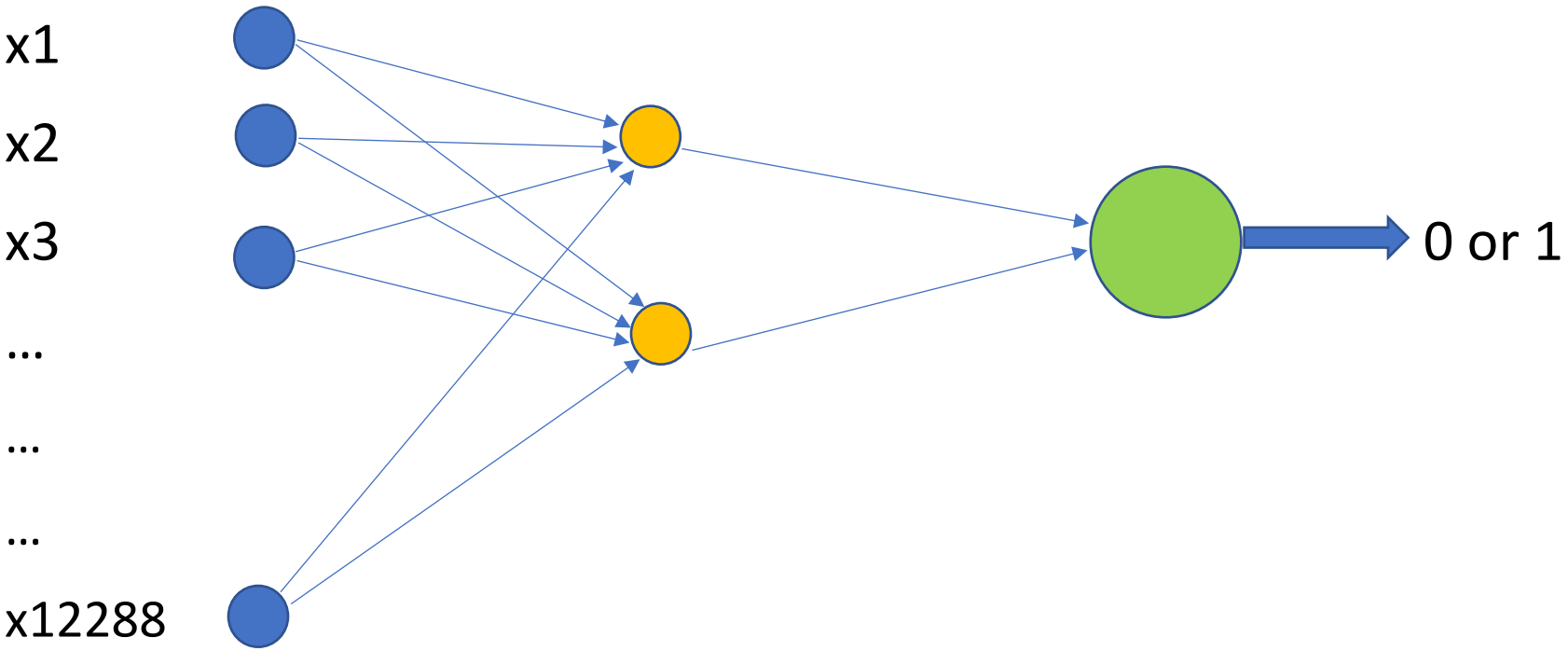
Grayscale ([0 ~ 255])

201	165	20	39	99	111	24	17	87	93
2	19	25	77	18	0	0	0	0	203
89	144	136	185	143	20	39	74	65	18
255	255	255	255	255	255	255	255	255	255
12	17	18	99	53	158	215	196	7	0
87	187	29	27	111	200	15	19	27	58
201	165	20	39	99	111	24	17	87	93

Color images (RGB)



of inputs = 12288


x1
x2
x3
…
…
…
x12288
0 or 1

- A single training example is represented by a pair (x,y) where x is an N dimensional feature vector and y is the label (zero or one).
- To put all the training examples into a more compact notation we are going to define a matrix (X).

